TPC-394: A Same-Count Origin-Uniformity Ladder for a Finite $c = 1$ Proxy

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Abstract

TPC-393 left a focused question: does its alternating-index origin-spread signal survive on another family after the normalization holdout? TPC-394 answers this with a response-blind eight-origin ladder at a common window length $N = 1024$. All four declared normalizations are stable at the one-percent origin-spread level for the all-plus control, while all four fail for the alternating-index law, with relative spreads between 0.084824884787110394 and 0.092863374514779065. The calibration-to- holdout mean transfer remains within a predeclared three-percent cap in all eight cells. The finite 0.64 spectral cap fails on all 32 all-plus rows and the Schur cap fails on no row. These are certified finite $c = 1$ proxy facts; they do not establish source validity, a growing origin-uniform estimate, arithmetic L^2, Route-A/Route-B closure, or a twin-prime theorem.

1 Question and claim boundary

TPC-393 found that a high-Q forecast anomaly from TPC-392 did not replicate, but that alternating-index origin spread remained above a one-percent diagnostic threshold. A direct same-count origin ladder is the smallest useful next test: it removes count-transfer from the primary statistic and asks whether the law-dependent signal survives a larger fresh family.

The status firewall is

$$\texttt{ARITHMETIC\_ADVANCE=NO}, \qquad \texttt{FIXED\_POWER\_CREDIT=0}, \qquad \texttt{FULL\_GATE\_B=OPEN}.$$

The official Session evaluator files are absent from this checkout. The local proof package and Bridge-B artifact are therefore fail-closed evidence of finite consistency only; they cannot declare an official Route-A or Route-B pass.

2 Finite proxy and frozen origin ladder

For $p \in (Q, 2Q]$, $H = 66$, and u, v in a finite interval, set

$$K_p(u,v) = p(p/Q)^2 \frac{H^2}{H^2 + (u-v)^2} \left(\mathbf{1}_{p \mid u-v} - \frac{1}{p-1} \right) \cdot \mathbf{1}_{u \neq v} \mathbf{1}_{p \nmid u} \mathbf{1}_{p \nmid v}.$$

The row geometry and signed matrices are

$$G(u) = \sum_p \sum_{v \in I} K_p(u,v)^2, \qquad M_\ell(u,v) = \sum_p s_\ell(p) K_p(u,v).$$

The fixed band retains block pairs at distance at most three. The candidate grid and selected origins are

$$a_j = 5000001 + 401j \quad (0 \le j < 41), \qquad j \in \{0, 5, 10, 15, 20, 25, 30, 35\}.$$

The first five origins are calibration and the last three are holdout. Every origin has $N = 1024$, partitioned into eight blocks of length 128. We use $Q = 8192$, the all-plus and alternating-index laws, and four fixed normalizations: local diagonal, pooled calibration scalar, current-origin scalar, and a first-calibration scalar frozen across origins.

Table 1: Origin-ladder summary. Each entry is a cell census over the two declared laws.

normalization	all-origin pass	max spread	transfer pass	spectral failures
local diagonal	1/2	0.0848248848	2/2	8/16
pooled train scalar	1/2	0.0928625707	2/2	8/16
origin scalar	1/2	0.0928633745	2/2	8/16
frozen train-1024 scalar	1/2	0.0928625707	2/2	8/16

For a cell, write $S(o)$ for the masked spectral diagnostic. The primary origin statistic is

$$R_{\mathrm{all}} = \frac{\max_o S(o) - \min_o S(o)}{\mathrm{mean}_o S(o)},$$

with a one-percent pass cap. The secondary holdout transfer error is

$$T = \frac{\mathrm{mean}_{o\in\mathrm{holdout}} S(o)}{\mathrm{mean}_{o\in\mathrm{calibration}} S(o)} - 1,$$

with a three-percent cap. Both cohort roles and caps were fixed before the current responses were read.

3 Certification protocol

The producer accumulates the prime shell in ascending order. An independent checker does not import the producer: it rebuilds the matrices in descending shell order and recomputes every row and aggregate. The canonical JSON certificate has 64 rows and 8 cells, exact TPC-393 parent hashes, and a rational 13-point anchor at $[5000001, 5000014)$ with $Q = 8$. The anchor checks positive geometry and exact symmetry for both laws. A 25-case mutation suite attacks hashes, roles, row census, summary fields, anchor data, and the claim firewall.

The producer and checker use float64 only for the finite matrix calculation; the opposite shell order is tolerated only at a small numerical replay tolerance. Ordinary and optimized runs are required to have identical outputs. The release Bridge-B checker locks all source, result, note, PDF, and compile-log artifacts.

4 Finite results

The complete panel has 64 rows and 8 cells. All four all-plus cells pass the one-percent origin rule. All four alternating-index cells fail it; their relative spreads, in local/pooled/origin/frozen order, are

$$0.084824884787110394, \quad 0.092862570673886716, \quad 0.092863374514779065, \quad 0.092862570673886591.$$

For comparison, the all-plus spreads in the same order are

$$1.5006633030031748 \times 10^{-5}, \quad 4.3100829567952307 \times 10^{-5},$$
$$2.2682851215503215 \times 10^{-5}, \quad 4.3100829568062604 \times 10^{-5}.$$

Thus the finite split is strongly law-dependent and is not removed by any of the four normalization choices.

The holdout-transfer cap passes in all eight cells. Its largest absolute error is 0.027694160160074421, so the observed obstruction is a same-count origin spread rather than a detected calibration-to-holdout level failure. The alternating/all-plus mean ratio is approximately 8.7×10^{-4}; this records finite cancellation in the selected proxy only.

The envelope diagnostics are asymmetric. The spectral cap fails in 32 of 64 rows, exactly the all-plus rows (8 per normalization), while no row fails the Schur cap. The Schur result is a finite diagnostic and not a growing Schur bound. Likewise, the spectral failure is a scoped obstruction to the chosen finite cap, not an asymptotic statement.

5 Interpretation and next clue

The strongest positive result is a response-blind, independently replayed same-count control: all-plus origin magnitudes are stable at roughly 10^{-5}–10^{-4} across the eight-origin ladder under all four normalizations, and all holdout-transfer cells pass. The strongest obstruction is the normalization-invariant alternating-index spread of about nine percent. This supports a finite law-dependent obstruction hypothesis, not a source-uniform theorem.

The reusable structure is a same-count origin ladder with a law control, frozen normalization panel, explicit calibration/holdout roles, exact parent provenance, reverse-shell replay, and mutation testing. The next clue is

`ROUND2_CLUE=TEST_C1_ORIGIN_CROSS_FAMILY_HOLDOUT`.

The next project should test whether this alternating origin obstruction transfers to another fresh affine family. Until a source-valid growing argument exists, no arithmetic power credit is assigned.

Reproduction

All source, proof, certificate, notes, and Bridge-B files are stored in the TPC-394 project directory. The README lists ordinary and optimized producer, independent checker, stress, and Bridge-B commands. The release requires `paper/main.pdf` and `paper/paper.pdf` to be byte-identical.